

Improving engineered biological systems with electronics and microfluidics

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Hybrid engineered biological systems integrate electronics and microfluidics with engineered biological components, such as microbes or cell-free DNA-based systems, to effectively sense, act upon and report on biological environments. As these engineered biological systems become essential in addressing challenges in healthcare, environmental monitoring, remediation and agriculture, this Review offers an in-depth discussion of their applications, critical design choices and challenges. Alongside an overview of the state of the field, we present a classification framework aimed at helping researchers make informed and optimized design decisions tailored to the intended biological application. Furthermore, we outline how the development of cyber-secure biological systems could enhance the security and functionality of engineered biological platforms. Last, we introduce a ‘Living Roadmap’ (<https://www.programmingbiology.org/csbs>) to dynamically reflect the field’s progress and support continuous monitoring of future advancements.

By leveraging engineered biological systems to sense, report and respond to environmental conditions, monitor human health or discover natural products, researchers can harness the inherent advantages of natural biological sensors^{1,2}. These advantages include robust performance under harsh environments, exceptional molecular sensitivity, cell-level specificity and self-replication capabilities^{1,2} (Fig. 1). Pure engineered biological systems use various microbial, mammalian and cell-free systems that are precisely engineered to perform specific functions, such as biosensing, logic-based control, gene therapy, chemical production and environmental remediation. These systems achieve their goals through the synthetic regulation of gene expression, including precise control over the timing and level of gene activity. Pure engineered systems are designed to operate exclusively within biological contexts, relying solely on cells or cell-free biological components without the integration of additional nonbiological technologies, such as microfluidics or electronics^{2,3}. However, despite important advances in engineered biology^{3–9}, the adoption of engineered biological systems in practical applications remains limited, primarily due to several challenges including difficulties in maintaining system functionality under diverse and dynamic real-world conditions¹⁰, the complexity of

scaling up from laboratory-scale to industrial-scale processes^{11,12}, the instability and lack of robustness of engineered traits¹³ and stringent regulatory requirements¹⁴. These challenges highlight the need for further research and development to bridge the gap between laboratory successes and practical, real-world applications. Moreover, the development of more advanced and precise tools for characterizing and understanding the complexities of biological environments is crucial for enabling the design of larger-scale, more reliable and robust systems constructed from biological components⁹.

Hybrid engineered biological systems integrate nonbiological technologies, such as microfluidics and electronics, to complement and extend the capabilities of pure engineered biological systems by leveraging the unique strengths and synergies of each domain (Fig. 1). Microfluidic devices allow for precise control of the microenvironment and fluid dynamics, enabling high-throughput screening and detailed analysis of biological processes^{15–17}. Additionally, microfluidic platforms substantially accelerate reaction kinetics while reducing the sample volumes required, thereby enhancing experimental efficiency and cost-effectiveness in high-precision analyses. The integration of electronic components facilitates real-time monitoring and regulation

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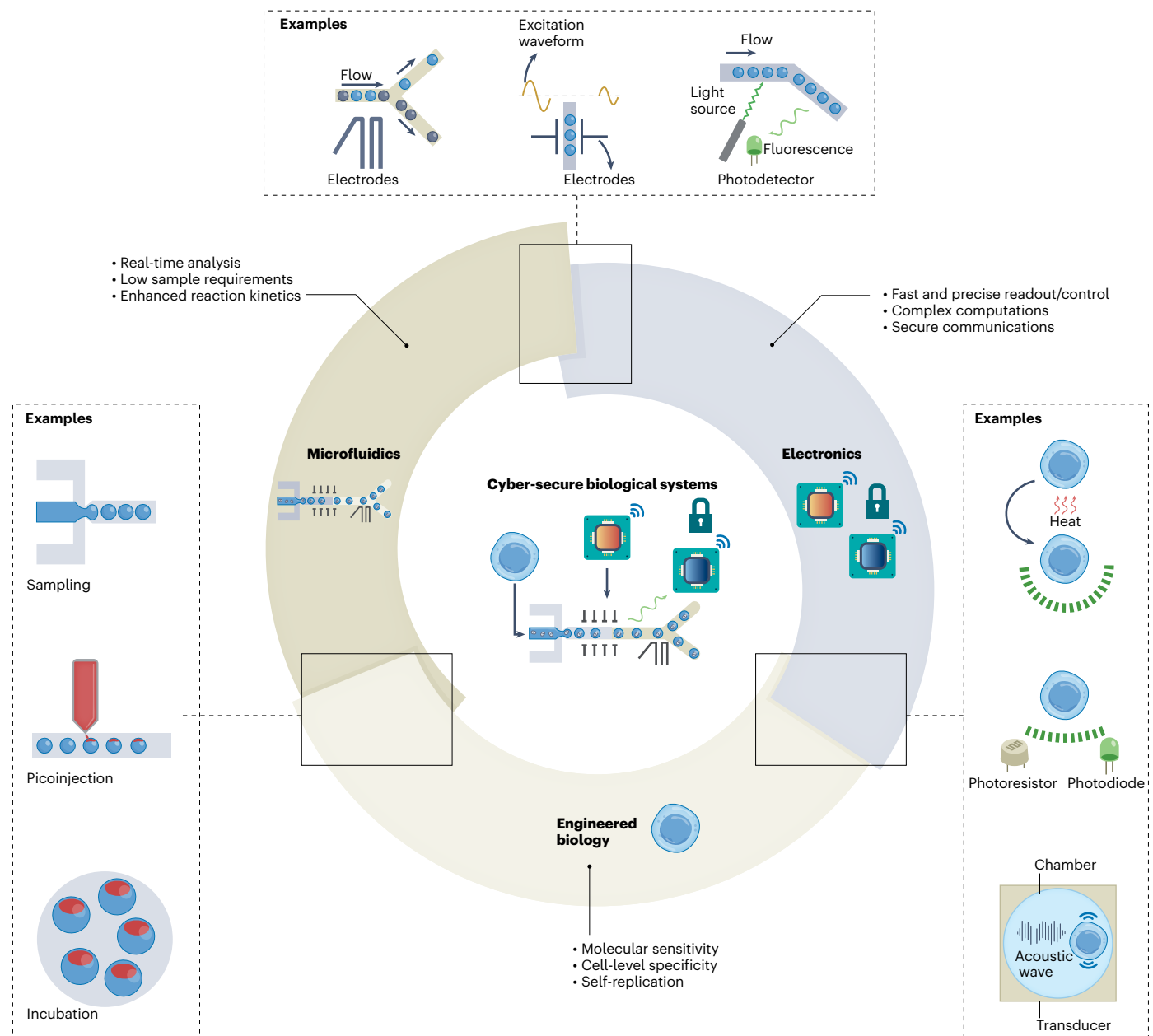


Fig. 1 | CSBS integrate engineered biology, microfluidics and electronics into a hybrid platform. Each domain contributes unique strengths, while their integration enhances real-world functionality and deepens our understanding

of complex biological processes. Of particular interest are the interaction points where each domain must be coupled to the other. The success of this coupling is crucial to overall system performance.

of biological processes through sensors and actuators, enhancing communication and computational functionality^{18–20}. For instance, integrating electronics into engineered systems facilitates precise readout and actuation at nanosecond timescales, thereby substantially improving the responsiveness and functionality of biological systems. Although these hybrid engineered systems hold promise for applications in remote or inaccessible areas, including environmental monitoring, medical diagnostics and therapeutic delivery, there are many considerations that researchers must take into account when using these systems.

Expanding on these capabilities, cyber-secure biological systems (CSBS) integrate biological components (for example, engineered living organisms) with advanced secure electronics (Fig. 1), culminating in a connected, autonomous network of biological internet of things (IoT) for real-world deployment. Engineered to communicate securely among themselves and with cloud-based platforms, these

systems facilitate data integrity while ensuring reliable connectivity in resource-constrained settings. By merging intrinsic biological self-correction and self-replication with intelligent data processing and control, CSBS continuously adapt, learn and evolve.

The integration of electronics and microfluidics with biological systems has been continuously evolving from the early 2000s to the present^{21–25} (Fig. 2). Moreover, potential future applications enabled by CSBS include smart agriculture, networked bioproduction and personalized healthcare. For instance, user-programmable engineered biological systems could support improved food crop growth and optimized production (<https://roadmap.ebrc.org>). These hybrid bio-electronics would monitor plant health in real time, track key metrics and provide feedback control to adjust water, nutrient delivery and greenhouse conditions. CSBS in networked bioproduction enables on-demand biomanufacturing of pharmaceuticals, food ingredients and chemicals. Autonomous bioreactor technologies can continuously

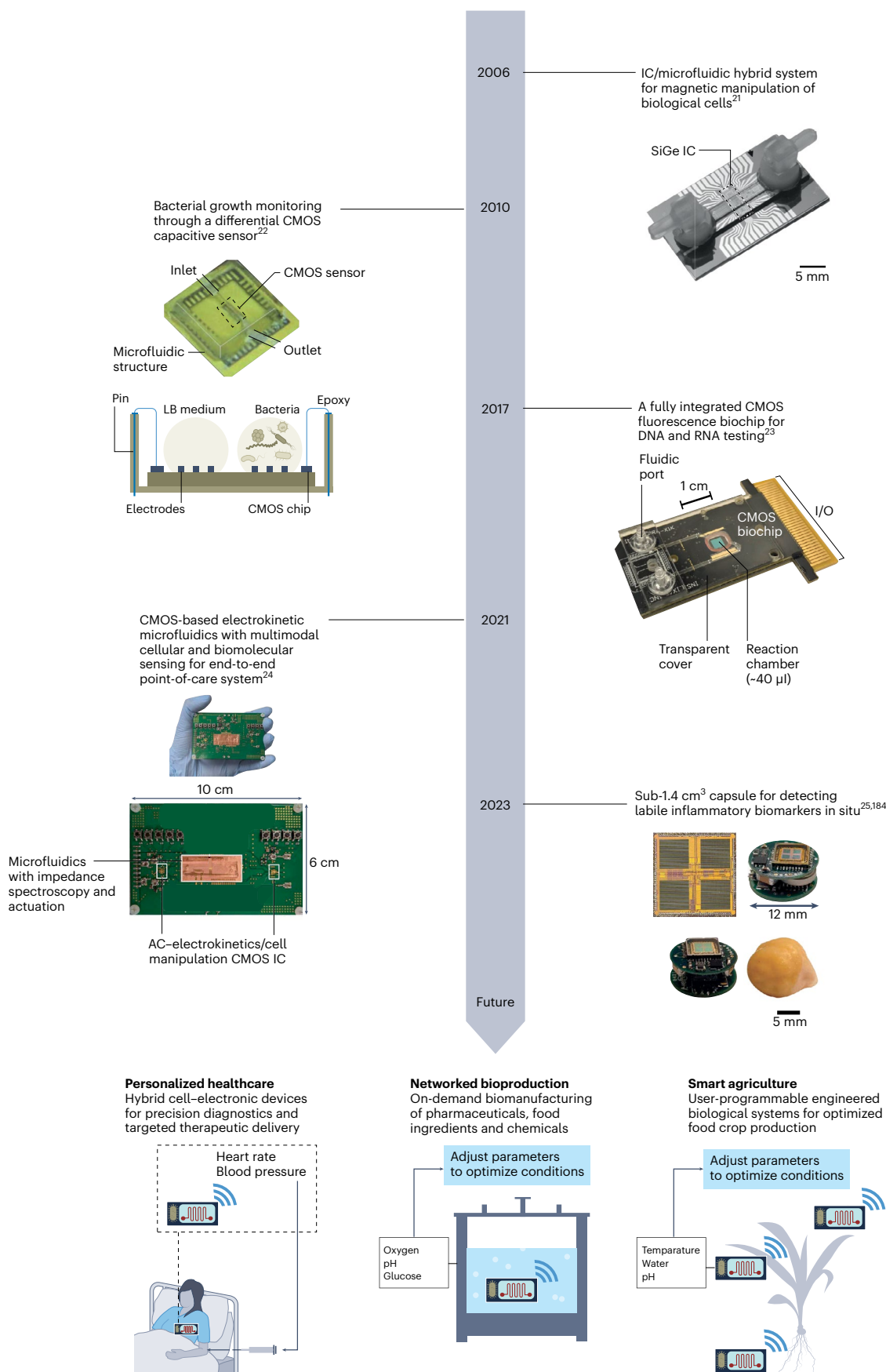


Fig. 2 | Timeline illustrating the integration of electronics, microfluidics and engineered biology into hybrid systems. Examples^{21–25} from the early 2000s to today highlight advances in integration, diverse sensing–actuation modalities and miniaturization, underscoring the evolution toward CSBS and

their promising future applications (<https://roadmap.ebrc.org>). AC, alternating current; CMOS, complementary metal–oxide–semiconductor; IC, integrated circuit; I/O, input and output; SiGe, silicon germanium.

monitor cell strain health, stress levels and productivity while dynamically adjusting parameters, such as pH, feedstock composition and temperature to optimize yield. These bioreactors can be engineered as microfluidic or nanofluidic platforms optimized for microgravity, making them well suited for space applications. Hybrid cell–electronic devices enhance personalized healthcare by enabling precision diagnostics through continuous, real-time, multimodal monitoring of metabolic and biochemical markers. Simultaneously, these wireless hybrid devices facilitate dynamic, feedback-controlled therapeutic interventions using engineered cells, tailoring treatments to individual patient needs. Another promising application is the deployment of wireless, hybrid engineered biological systems for real-time monitoring of built environments and natural habitats (<https://roadmap.ebrc.org>).

This Review provides an overview of how engineered biological systems can work together with microfluidics and electronics to address practical challenges and unlock these new applications. We focus primarily on hybrid biological systems that integrate engineered living organisms or cell-free components with electronics for complex computations, communications and real-time feedback control, housed within microfluidic environments. We aim to bridge these complementary disciplines to foster effective cross-disciplinary collaboration between biologists and electronics and microfluidics researchers. A use case of this work would be for biologists to identify electronics or microfluidics that might augment their biological systems' precision, efficiency or applicability. Similarly, a microfluidic or electrical engineer could identify biology that could increase the power of their platform. This work would allow those individuals to recognize the key technologies and researchers in those areas, thereby creating new collaborations, research opportunities and actual deployed systems. By combining insights from these diverse fields, we aim to catalyze the development of next-generation biological systems that would be unattainable through isolated approaches. We provide guidelines on design considerations, performance trade-offs and system classifications, offering a roadmap for optimizing integration across complementary domains. We discuss current challenges and identify emerging research opportunities to advance engineered biological systems, focusing on improving their use in practical applications, including healthcare, environmental monitoring, bioremediation and agriculture. Finally, we describe a 'Living Roadmap' accessible via <https://www.programmingbiology.org/csbs> that reflects the current state of the field^{2,5,13,16–171} while enabling seamless tracking of future advances and updates in this rapidly developing area of research.

Modalities of microfluidic designs for engineered biological systems

Microfluidic devices offer precise and dynamic control over environmental parameters, such as temperature, pH, nutrient gradients, gas concentrations and chemical exposure, which is particularly advantageous in engineered biological systems where regulation of cell behavior and gene expression is crucial. Additionally, these platforms facilitate high-throughput experimentation by manipulating minute fluid volumes, thereby supporting thousands of parallel assays (Fig. 1). Operating at microliter and nanoliter scales reduces the consumption of expensive reagents, enzymes and cells, while the confined volumes promote faster diffusion, mixing and reaction kinetics than those in conventional bulk experiments (Fig. 1). Furthermore, microfluidic systems offer the ability to isolate and monitor individual cells, providing valuable insights into cell-to-cell variability and stochastic gene expression, and they support co-culture studies involving multiple organisms or synthetic microbial consortia.

For deployed hybrid engineered biological systems, microfluidic housing establishes a well-controlled artificial environment that not only ensures optimal interfacing between biological and electronic components but also regulates critical parameters, such as fluid flow,

temperature and chemical exposure, to shield sensitive elements from external disturbances. This controlled setting in a confined space minimizes the risk of signal interference and component degradation, localizing potential malfunctions and thereby enhancing overall system reliability and user safety.

Diverse modalities of microfluidic designs used in engineered biological systems can be categorized based on the paradigm of operation, fluid flow mechanism, liquid flow control, materials used in fabrication and the degree of customization and automation in the design process (Fig. 3). Microfluidic device paradigms can be classified into passive, electrowetting, droplet-based and continuous flow systems. Passive systems use capillary action and surface tension to drive fluid movement without external power, common in point-of-care diagnostics and simple biochemical assays¹⁷². Electrowetting systems employ electric fields to alter the wettability of surfaces, enabling precise control of liquid droplets, used in digital microfluidics for sample preparation, mixing and analysis^{173,174}. Droplet-based microfluidic systems, which exploit magnetic fields or immiscible fluid interfaces to generate and manipulate discrete droplets, are ideal for high-throughput screening, single-cell analysis and compartmentalized reactions^{175,176}. A notable example is the microfabricated emulsion generator array⁴⁹, capable of producing up to 3.4 million nanoliter-sized droplets per hour, thereby enabling detailed genetic analysis at the single-cell level. In this system, individual cells are encapsulated within droplets alongside microbeads functionalized with primers targeting specific genes. This configuration facilitates precise multiplex polymerase chain reaction analysis of genetic variations within complex cell populations. Moreover, the microfabricated emulsion generator array system incorporates hybrid glass–polydimethylsiloxane diaphragm micropumps for controlled droplet formation and efficient encapsulation. This design enables high-throughput digital detection and quantification of specific cell types, exemplified by the identification of pathogenic *Escherichia coli* amid a background of normal K12 cells. By contrast, continuous flow microfluidic systems rely on external pumps and fluid dynamics to maintain a steady flow through microchannels, making them suitable for applications requiring continuous processing, such as chemical synthesis and cell culture¹⁷⁷.

Fluid flow mechanisms within these devices include capillary action in passive systems, electric fields in electrowetting and electrophoretic systems, magnetic fields in droplet-based systems and pressure-driven flow in continuous flow systems. Liquid flow is typically controlled by valves, electrodes, magnets or pneumatic pumps¹⁷⁸. Valves are mechanical or microfabricated components that regulate fluid flow, electrodes are used in electrowetting and electrophoresis for precise control, magnets steer magnetic beads or ferrofluids in droplet-based systems, and pneumatic pumps generate pressure-driven flow in continuous and pneumatic systems.

The materials used for constructing microfluidic devices¹⁷⁹ include glass, which offers high chemical resistance, optical transparency and mechanical stability; polymers, such as thermoplastics and polydimethylsiloxane, which provide flexibility, ease of fabrication and cost-effectiveness; organic fibers like paper, enabling low-cost and disposable devices; and hydrogels, providing a biocompatible and tunable environment ideal for tissue engineering and biomolecular assays.

Currently, most microfluidics are designed using generic software that primarily focuses on the physical layout, often overlooking functional and performance requirements. To address this limitation, an approach using computer-aided design tools, as suggested by recent studies^{180,181}, would enable the creation of a standardized library of microfluidic components based on functional specifications. These components could be configured and parameterized according to performance criteria, facilitating rapid design iterations, reuse of specifications as fabrication capabilities evolve and formal verification of design specifications. Designs can be fully customized to meet specific application requirements for maximal flexibility and functionality.

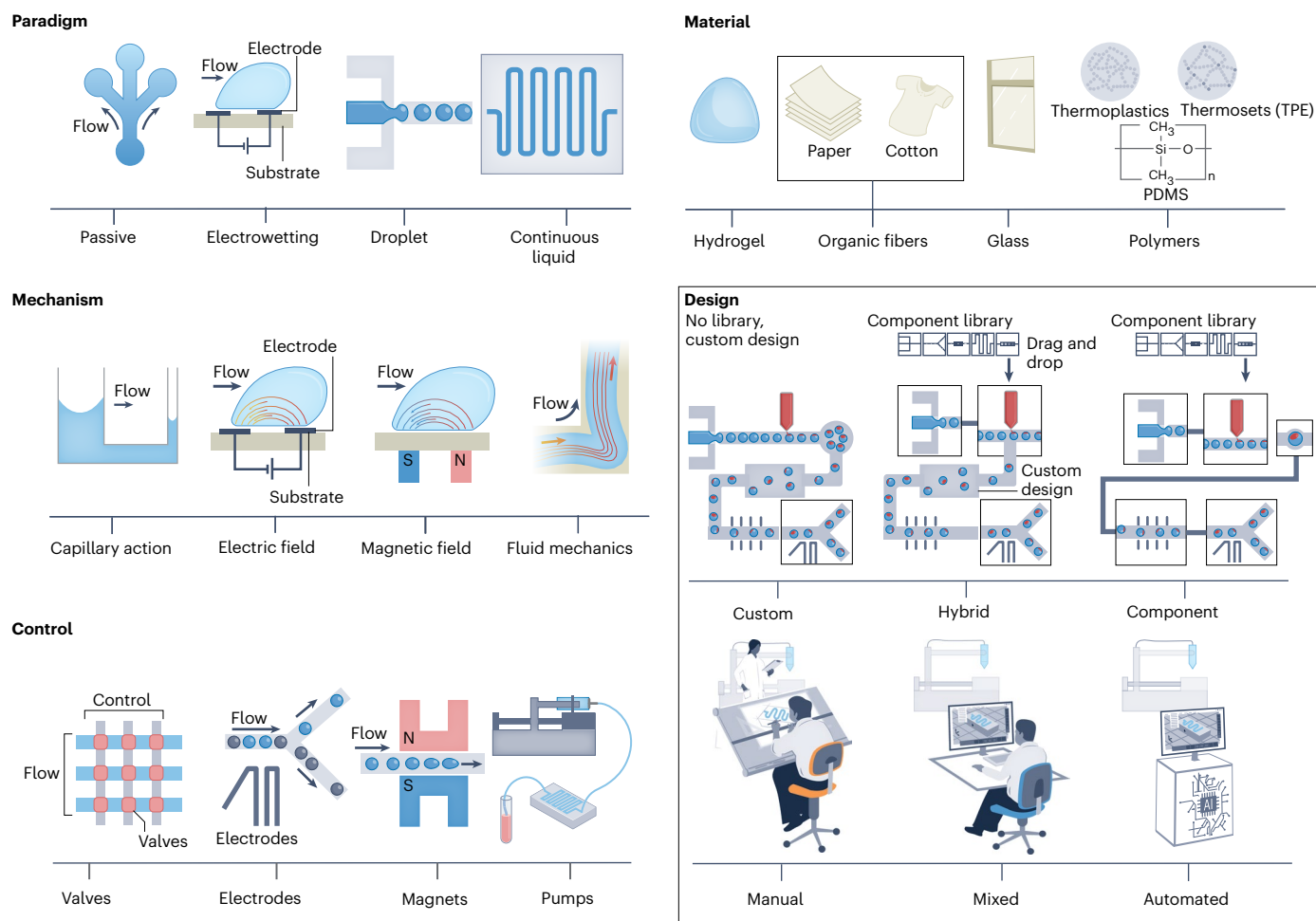


Fig. 3 | Overview of microfluidic technologies in hybrid engineered biological systems. Technologies are categorized into (1) microfluidic device paradigms, including passive, electrowetting, droplet-based and continuous flow systems, (2) mechanisms of fluid flow, such as capillary action, electric and magnetic fields and fluid mechanics, (3) methods for liquid flow control, featuring valves, electrodes, magnets and pneumatic pumps, (4) materials for fabrication, encompassing glass, polymers, organic fibers and hydrogels and (5) levels

of design customization and automation, from fully customized designs to modular component libraries and hybrid designs. These are not exhaustive or meant to comment on the advantages of one technology over the other. However, if one were to select technologies from each category, it would begin to define which applications are possible, what biological systems are necessary and what electronic components may be needed. PDMS, polydimethylsiloxane; TPE, thermoplastic elastomer.

Alternatively, modular component libraries with standardized, interchangeable parts can simplify design and assembly, promoting scalability and rapid prototyping. Hybrid designs, combining bespoke elements with modular components, balance customization with the efficiency of using prefabricated parts and are facilitated by advanced computer-aided design tools.

The inherent scalability of modular design is exemplified by the development of low-cost, reconfigurable microfluidic systems that effectively address the complexity and limitations associated with conventional fabrication techniques. By employing modular, discrete components, researchers can assemble complex three-dimensional (3D) microfluidic networks optimized for common passive functions, such as junctions, mixers and splitters⁴⁸. These systems facilitate tunable mixing, reconfigurable droplet generation and in situ monitoring of droplet production, enabling the rapid assembly, modification and optimization of devices tailored to specific experimental requirements. Building upon these advancements, programmable multipurpose microfluidic platforms, such as those featuring an addressable grid of 95 nanoliter-volume chambers for multiparameter analysis of microbial communities⁵², exemplify the precision and adaptability afforded by modular design. Their functionality is achieved through the integration of droplet-based sample compartmentalization with dynamically

reconfigurable flow-routing control, enabled by the incorporation of integrated microvalves. Consequently, these platforms provide robust, versatile tools for engineering biological systems, allowing complex analyses while minimizing cross-contamination and supporting the selective recovery of cells.

Predictive modeling and performance analysis of small-scale fluid dynamics in microfluidic environments pose notable challenges¹⁸². These predictions are needed to abstract the design process to high-level specifications. An alternative approach could employ the design of experiment methodology to intelligently sample the design space, build prototypes, measure performance and use machine learning (ML) and artificial intelligence (AI) to create models based on empirical data rather than purely mathematical equations⁵⁵. The integration of AI and ML could enhance design accuracy and efficiency.

Accessibility remains a considerable challenge for microfluidic devices due to their often costly fabrication processes, specialized expertise and complex setup and maintenance requirements. However, low-cost approaches using materials like thermoplastics and techniques such as 3D printing offer promising alternatives that balance cost, production time, accuracy and scalability¹⁴⁵. For instance, a method involving 3D-printed molds and silver conductive paint has been developed to create dielectrophoretic microfluidic devices for

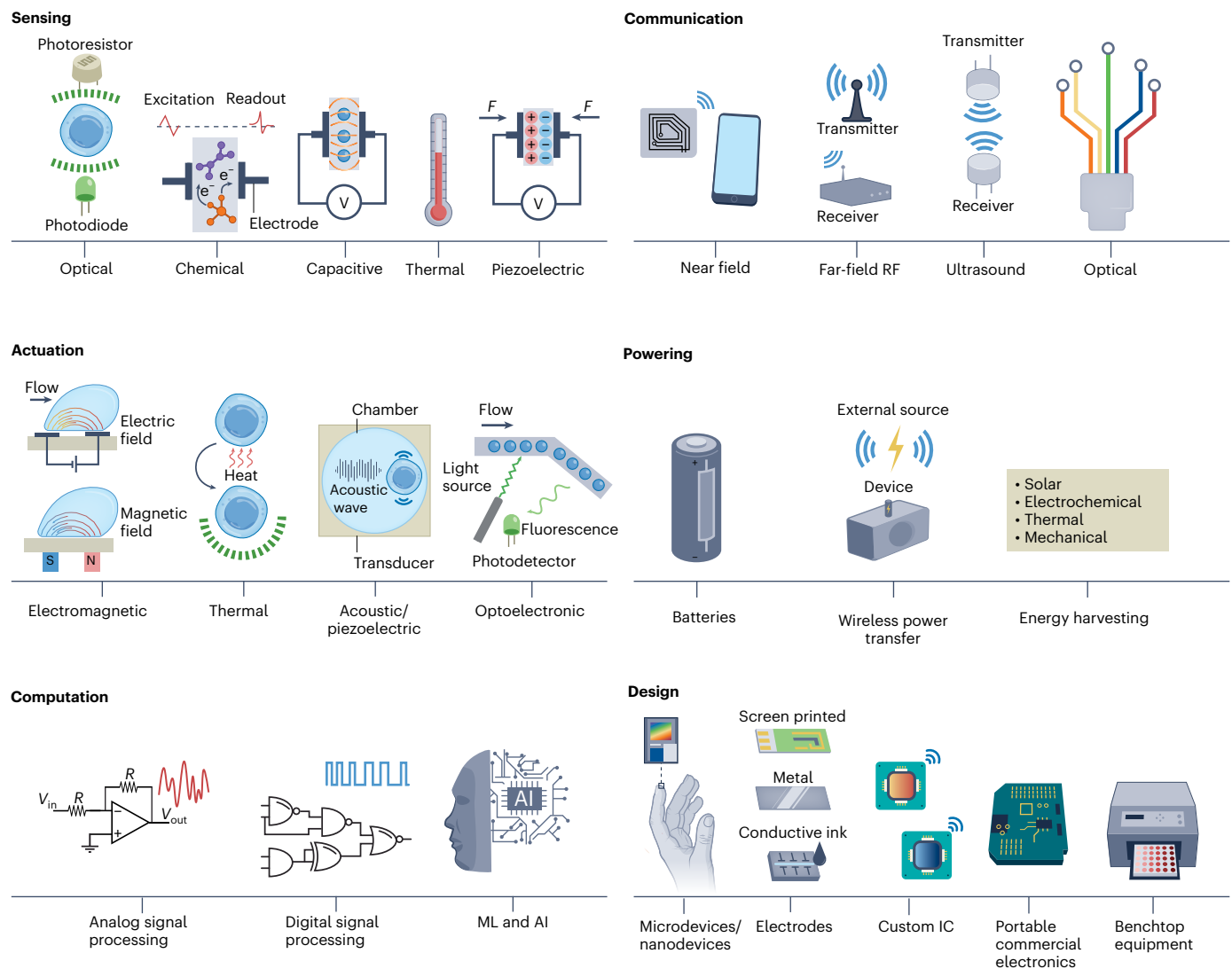


Fig. 4 | Overview of electronics in hybrid engineered biological systems. They are categorized into six key paradigms: (1) sensing modalities, including optical, electrochemical or chemical, capacitive, thermal, and piezoelectric techniques, (2) actuation modalities, covering electromagnetic, thermal, acoustic or piezoelectric, and optoelectronic methods, (3) computation techniques, using analog and digital signal processing, ML and AI, (4) communication methods, ranging from near-field and far-field RF to ultrasound and optical, (5) powering

technologies, featuring battery advancements, wireless power transfer and energy harvesting and (6) design principles and form factors, including microdevices, nanodevices, electrodes, custom ICs, portable commercial electronics and high-precision benchtop equipment. Systems integrate these paradigms to create tailored solutions, combining sensing–actuation, computation, communication and powering techniques to meet specific application requirements in biological environments.

particle separation⁴⁰. This approach substantially reduces production costs and time, while the use of silver conductive paint provides an economical and straightforward method for forming electrodes. These devices employ a nine-pole comb electrode configuration to generate an electric field, exemplifying the integration of more accessible and cost-effective manufacturing technologies. By enabling the development of efficient and functional microfluidic devices without complex and expensive fabrication methods, these innovations broaden accessibility for researchers and expand the range of potential applications.

Despite these advances, biocompatibility challenges remain, particularly in selecting suitable materials, managing internal and external environmental conditions, controlling biological timescales and ensuring the reproducibility of results¹⁸³. Addressing these issues could further enhance the efficiency and accessibility of microfluidics, fostering innovation in the design and application of hybrid engineered biological systems.

Electronic modalities for engineered biological systems

The integration of electronic computation and communication components strengthens the secure processing, storage and sharing of data within engineered biological systems (Fig. 1). This facilitates the development of sophisticated hybrid devices that support continuous, real-time monitoring of both the deployment environment (for example, toxic chemicals in wastewater, molecular disease biomarkers in the human body) and biological viability (for example, cell density, electrical impedance of the cell membrane). Electronic augmentation underpins closed-loop feedback mechanisms that dynamically regulate biological conditions to ensure optimal performance even in adverse or harsh environments (for example, temperature and pH regulation). Furthermore, the stark contrast in time constants, in which biological processes operate on the scale of minutes to hours and electronic components function within nanoseconds to microseconds, presents unique integration opportunities. Leveraging this mismatch can help

tailor hybrid systems to a diverse range of application requirements, optimizing both responsiveness, power consumption and resource use.

Key elements in electronic design are classified according to sensing and actuation modalities, computational domains, communication and powering strategies as well as design miniaturization, customization and integration levels (Fig. 4). Collectively, these factors determine the efficiency and stability of bidirectional information transfer between electronic and biological systems.

Several technical challenges must be addressed to integrate engineered biological systems with electronic devices for enhanced real-world applicability. Current information transfer mechanisms for interfacing these systems are limited in versatility, necessitating the development of orthogonal mechanisms for both sensing and actuation¹⁸⁴. Sensing modalities are categorized into optical, electrochemical or chemical, capacitive, thermal, and piezoelectric, whereas actuation modalities are categorized into electromagnetic, thermal, acoustic or piezoelectric, and optoelectronic. For example, an integrated circuit (IC) with an overlaid microfluidic layer forms a hybrid system for the magnetic manipulation of bead-bound biological cells (Fig. 2). This IC technology, featuring a microcoil array, enables the simultaneous manipulation of multiple individual cells with precise spatial control²¹. Modern silicon IC processes can integrate multiple modalities into a single complementary metal–oxide–semiconductor (CMOS) electronic chip. This multimodal integration facilitates cross-validation of target detection and calibration, ensuring the generation of reliable data and efficient signal transduction between the electronic components and engineered biological systems. For example, combining cell density measurements with luminescence sensing is crucial for understanding the status and life cycle of engineered biological cells, which is vital for sensor calibration. Various CMOS-based electrochemical impedance spectroscopy and capacitive-sensing architectures have been developed for cellular and biomolecular sensing and can also be used to characterize cell populations and monitor living organism growth^{22,24,91,95} (Fig. 2). Sensing and actuation modalities must operate efficiently within constraints of energy consumption and device size while enhancing the interaction between engineered biological systems and electronic devices for seamless integration and functionality in practical applications.

Recent research highlights the advanced capabilities of fully integrated CMOS electronics in multiplexed measurement and control, enhancing both sensing and actuation functions. For instance, a fully integrated CMOS-based microfluidic system showcases precise fluid flow control via alternating current electroosmosis, eliminating the need for bulky pressure-driven systems such as syringe pumps (Fig. 2). This platform also performs advanced cell manipulation and label-free protein detection through an impedance spectroscopy receiver array with 16 hardware channels²⁴. Another study¹²⁰ introduces a multifunctional CMOS biosensor array with 4,096 working electrodes and 12,288 photodiodes for versatile applications, including dielectrophoretic actuation, impedance sensing and optical imaging, thereby enhancing real-time biosensing for the detection of low-concentration analytes.

Even optimized signaling from engineered biological systems tends to be very weak (for example, optical signal response with a strength of 20–200 photons per second per cell¹⁸⁵) and has traditionally been quantified using bulky, power-hungry laboratory setups that provide high precision and sensitivity (for example, plate readers and gel imagers for optical signals). For field deployment, where energy efficiency is a critical constraint, it is crucial to achieve high sensitivity and multiplexed measurements with a wireless portable electronic component that can acquire, encode and communicate weak signals. Recent advancements in miniaturized device technology, such as CMOS fluorescence sensors^{20,23,113,114} (Fig. 2), address some of these challenges. CMOS fluorescence sensors integrate on-chip optical filters, photodetectors and computation circuits to enable dynamic bidirectional communication with living cells. By leveraging standard

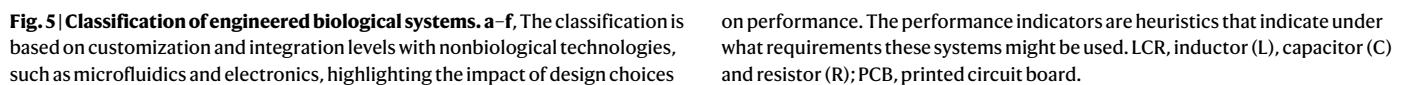
CMOS technology, the traditionally bulky optical filtering components are miniaturized using metal and dielectric layers, while sensitivity is enhanced via circuit techniques, such as correlated double sampling, which mitigates hardware offset and low-frequency noise. Additionally, innovations such as ingestible bacterial–electronic systems^{25,102} that integrate biosensor bacteria with ultra-low-power CMOS luminescence readout technology offer real-time, noninvasive biomolecular detection within the gastrointestinal tract (Fig. 2). The continued miniaturization and integration of key components, such as photodetectors and wireless transmitters, onto a single CMOS chip highlight the progress toward highly compact, application-specific biosensor systems tailored for field-based and in situ diagnostic applications.

Advanced sensing and computation techniques in the analog or digital domain, potentially leveraging ML methods, are required to accurately detect and process multiple signals simultaneously while extracting essential signal features with minimal power consumption. This includes integrating low-power, low-noise sensor technologies¹⁸⁶ with optimized data acquisition methods and efficient signal-processing algorithms¹⁸⁷. Ensuring data security necessitates robust encryption, secure authentication protocols and data anonymization techniques to protect against unauthorized access and data manipulation¹⁸⁸.

Real-time information transfer from hybrid engineered biological systems is enabled by wireless communications. However, effective transmission in deployed settings presents unique technical challenges, involving the selection of appropriate communication frequencies and modulation schemes while satisfying application needs for communication range and signal directionality. Communication modalities can be classified into near-field and far-field radio frequency (RF), ultrasound and optical categories. Far-field RF communication faces a fundamental trade-off between antenna size and far-field radiation efficiency, as indicated by the Wheeler–Chu limit¹⁸⁹. Near-field communication decouples the antenna size from the operating frequency, which simplifies the design by allowing lower-frequency operation with power savings. However, this comes at the expense of limiting communication to shorter distances and signal availability in only certain orientation axes. Stringent energy and size constraints intensify these challenges, necessitating the development of robust, low-power communication methods and protocols to ensure reliable, secure and efficient data transfer outside controlled laboratory environments.

Efficient power supply is essential to extend operational lifespan, ensure biocompatibility and enhance communication capabilities. These needs can be addressed by implementing innovative power management strategies¹⁹⁰, integrating energy harvesting solutions¹⁹¹ and optimizing the power consumption of electronic components, including sensors, actuators, computing units and communication modules¹⁹². Commonly used batteries limit device miniaturization and create safety issues due to toxicity risk^{159,189}. Remote RF-based powering, also known as wireless power transfer, faces practical efficiency limitations due to substantial power loss over long distances and in challenging environments with high path loss, particularly in deep tissue for medical devices or in water for environmental sensors. The need for large antennas and issues related to transmitter–receiver orientation and alignment hinder widespread adoption^{159,193}. Popular energy harvesting techniques that can be applied to power the electronic components of hybrid engineered biological systems consist of galvanic couples, which use electrochemical reactions¹⁶⁰, biocatalysts that convert biological reactions into electrical energy¹⁸⁹, microbial fuel cells that generate power through microbial metabolic processes^{194,195}, solar power through photovoltaic cells^{196,197}, thermal-based approaches using thermoelectric generators to capture temperature differences¹⁹⁸ and vibration-based approaches employing piezoelectric materials to convert mechanical energy into electrical power¹⁹⁹.

Engineered biological systems interface with electronic designs in various form factors and levels of integration and customization.



underscore the potential for real-time, bidirectional communication between electronic devices and biological systems, thereby creating new opportunities for the development of advanced biosensing technologies, biocomputational platforms, molecular diagnostic devices and programmable therapeutic interventions.

To realize this potential in real-world deployment of hybrid engineered biological systems, it is essential to improve their robustness and reliability. This involves several key aspects of device design that bridge different research areas, such as microfluidics, electronics and material science. Environmental exposure to moisture, dust and temperature extremes can degrade system performance, necessitating the development of robust encapsulation and sealing techniques to protect systems against harsh conditions²⁰². It is essential to use biocompatible materials and surface treatments to prevent biofouling or inflammatory responses, particularly for medical devices that must interact with biological tissues long-term. Mechanical stresses demand the use of flexible, durable materials, which should be rigorously tested to ensure long-term reliability²⁰³. By incorporating redundancy through highly parallelized microfluidic or electronic components, system reliability can be greatly improved and failure risks can be reduced, ensuring that they perform effectively in real-world conditions¹²⁶. Last, the seamless

BOX 1

The classification of engineered biological systems into six distinct categories

BioPOLOS: biological pure optical laboratory-scale off-the-shelf systems

This class encompasses pure engineered biological systems designed for laboratory development, devoid of microfluidics or electronic components^{2,5,26–35}. Their performance is evaluated using precise benchtop instruments, with plate readers being the preferred choice for analyzing optical properties, such as absorbance, fluorescence and luminescence. These measurement techniques are versatile and applicable across various biological chassis, provided that the sample's optical characteristics align with the detection capabilities of the instruments.

BioHELOS: biological hybrid electrical laboratory-scale off-the-shelf systems

This class focuses on hybrid engineered biological systems that use microfluidics for controlled manipulation of small liquid volumes in laboratory development^{36–45}. Their performance is typically evaluated using precise electrical benchtop instruments, such as potentiostats and LCR meters. LCR meters assess cell density and growth through impedance measurements, providing valuable data on cellular properties and behavior. Potentiostats are used to analyze redox reactions within living cells by precisely controlling the potential and measuring the current in electrochemical systems. This approach allows for a detailed investigation of electron transfer processes and metabolic activity at the cell interface, providing essential insights into cellular function, bioenergetics and redox homeostasis.

BioHOLOS: biological hybrid optical laboratory-scale off-the-shelf systems

This category involves hybrid engineered biological systems analyzed using optical techniques within microfluidic setups^{16,43–73}. Key benchtop instruments include flow cytometers, which analyze cells based on morphology, size and protein expression and incorporate droplet microfluidics for efficient cell sorting. Fluorescence detectors equipped with fiber optics enhance sensitivity, enabling real-time monitoring of biomolecular interactions within microfluidic environments. Photon counters are used to quantify light emission from bioluminescent or fluorescent signals, crucial for evaluating engineered biological systems such as biosensors or gene circuits.

BioHEMOS: biological hybrid electronic miniaturized off-the-shelf systems

This class integrates hybrid engineered biological systems with portable commercial electronics to facilitate data generation, processing and wireless communications^{74–82}. Despite the microfluidic and electronic functions not being integrated into a single device or highly customized for specific deployment scenarios, these systems are designed for field deployment owing to their portable nature. For instance, off-the-shelf electronic systems enable various electrochemical sensing techniques, such as electrochemical impedance spectroscopy, cyclic voltammetry, amperometry, potentiometry, ion concentration measurements and temperature sensing.

BioMICS: biological multicomponent integrated customized systems

This category includes hybrid engineered biological systems that integrate customized electronics and microfluidics^{83–97}. Unlike the BioHEMOS, which employ portable off-the-shelf electronic components, these systems feature custom-designed CMOS electronics specifically designed to meet application requirements. This customization facilitates precise interaction between the engineered biological response and electronic readout or control mechanisms. In these systems, microfluidic, electronic and biological components function both distinctly and collaboratively. However, they are not designed for complete integration as stand-alone systems suitable for remote applications.

(Sec)BioFICS: (secure) biological fully integrated customized systems

These hybrid systems leverage custom CMOS electronics to enhance reliable data generation and wireless communication, capitalizing on the specialized sensing and reporting functions inherent to engineered biological systems similar to the BioMICS. These systems operate within tailored microenvironments, such as microfluidic housings, facilitating optimal integration between electronic and biological elements. In contrast to the BioMICS, this category underscores fully integrated, highly customized solutions suited for diverse applications and field deployment^{20–25,98–137}. For instance, certain configurations incorporate physical layer security measures in electronics, effectively managing constraints, such as size and energy efficiency, to ensure robust and reliable performance across varied operational scenarios.

integration of microfluidic and electronic components²⁰⁴ with existing engineered biological systems requires standardized interfaces, modular designs and comprehensive compatibility testing.

Classifying engineered biological systems

We propose a classification that divides engineered biological systems into six categories, providing a standardized framework for the integration of hybrid devices in future biological applications (Fig. 5). This structured approach enables systematic performance trade-off analysis, supporting more informed and optimized design decisions tailored to specific biological applications. We base our classification on 121 papers^{2,5,16,20–137} published from the early 2000s to 2024, describing systems that differ in integration level, cost, feasibility and precision, depending on their intended use case, technologies

employed and their performance characteristics (Box 1). By default, these systems are considered unsecured. Future applications of hybrid engineered biological systems will likely demand integrated security measures to ensure reliable functionality and performance in real-world environments, safeguard user safety and protect data, enabling them to operate as a securely connected 'biological' IoT network (Box 2).

The different categories of engineered biological systems exhibit both commonalities and distinct variations across five key performance metrics (Fig. 5 and Supplementary Fig. 1). First, systems differ in their integration level. The level of integration changes as more functions are combined into a single device. Integration can improve the deployment abilities, reproducibility, size and ease of use. The trade-off is that devices must be produced at a volume

BOX 2

Toward cyber-secure biological systems

Hybrid engineered biological systems, such as BioHEMOS, BioMICS and BioFICS, leverage electronic augmentation to enable real-time data collection, analysis and control via wireless connectivity. By transforming biological functions into smart, interconnected devices, these systems form wireless bioengineered sensor networks that operate analogously to a biological IoT network. While this convergence offers advanced capabilities like continuous monitoring, remote diagnostics and automated interventions, it also introduces critical security risks^{188,216}. The expanded connectivity exposes these systems to cyberattacks, which can compromise data integrity, device functionality and overall system resilience.

In the biomedical domain, wireless devices integrating hybrid engineered biological components for therapeutic delivery or diagnostics become vulnerable to hacking. Such breaches can expose sensitive patient data or disrupt device operations, potentially posing life-threatening risks²¹⁷. Embedding robust security measures within these devices, especially as they increasingly connect to hospital networks and patients' mobile devices, is essential for maintaining uninterrupted diagnostics, treatments and patient safety.

Similarly, the deployment of hybrid bioelectronic systems in agriculture introduces vulnerabilities. A coordinated cyberattack targeting soil-monitoring systems could manipulate data streams, leading to improper irrigation schedules, nutrient mismanagement or inaccurate pesticide applications^{218–220}. These disruptions can harm crop health and reduce yields. As interconnected biological–electronic networks become integral to smart agriculture,

safeguarding them is crucial to prevent agricultural losses, supply chain disruptions and food insecurity. Robust countermeasures are necessary to protect both biological information and operational integrity across these systems, ensuring resilience and reliability in healthcare, agriculture and other real-world deployment scenarios.

To address these challenges, CSBS have emerged as a comprehensive solution. By integrating engineered biological components with advanced technologies, such as secure custom CMOS electronics, CSBS enhance resilience and security in real-world applications. Furthermore, these systems are engineered to evolve continuously by integrating inherent biological self-correction and self-replication with smart data processing and control mechanisms. This broad category encompasses specific classes like SecBioFICS and other future hybrid biological systems. Given the tight constraints on energy, computation and size, securing these systems cannot rely solely on conventional cryptographic mechanisms. Instead, security must be embedded at the forefront of the design process, addressing unique challenges at the intersection of engineered biological systems and cybersecurity. For example, upon detecting anomalies, these systems can shift to a secure mode that enhances protection and robustness at the expense of performance. Once the anomaly is resolved, the system reconfigures to a low-power mode, optimizing performance with moderate security. This dynamic adaptation of security measures to environmental conditions conserves power compared with fixed-security approaches.

to justify nonrecurring engineering costs. Second, systems vary in field deployability, which increases when they are fully integrated, require minimal physical interaction after deployment and use wireless communications. The third metric to consider is reusability. Reuse improves for in-laboratory systems used for development, whereas hybrid systems designed for deployment may be single use. Fourth, different applications of engineered biological systems affect cost-effectiveness. Costs increase for systems in the development stage requiring precise benchtop equipment. Conversely, fully integrated hybrid systems using custom CMOS electronics are optimized for cost-efficiency in field deployment. These systems, which use custom-designed CMOS electronics, involve semiconductor chip fabrication and rigorous testing. Nonrecurring engineering costs for chip fabrication dominate in low-volume production (for example, 100 research chips). However, as production volume increases (for example, 1 million production chips), fixed costs are spread over more units, reducing the per-unit cost through economies of scale. Consequently, fully integrated systems become more cost-effective for large-scale production than alternatives. Last, systems vary in their level of precision. Measurements using expensive benchtop equipment are more precise than those conducted with cost-effective portable CMOS electronics. Precision is higher in benchtop equipment because it can use more powerful processing components without resource constraints. By contrast, portable CMOS electronics face precision limitations due to design trade-offs for compactness and power efficiency, such as limited processing power and storage capacity.

To help researchers to choose which system would be most beneficial for their research project, we constructed a decision tree (Fig. 6) that can also be found at <https://www.programmingbiology.org/csbs>. While this decision tree focuses on illustrative examples that highlight

the reasoning behind different decision pathways, we provide a more prescriptive version of the diagram in Supplementary Fig. 2.

Future directions

The classification of electronic and microfluidic biological systems illustrates how technology interacts with biological systems in various exciting ways. The classification criteria not only highlight current progress but also pave the way for future innovations in the CSBS design ecosystem. We anticipate that the evolving landscape will feature a strategic division of roles, with sensing functions primarily managed by engineered living organisms or cell-free systems, whereas complex computations, secure communication and real-time feedback are conducted by secure custom ICs, interfacing with these biological components through advanced microfluidic systems. This approach harnesses the reliability and sophisticated communication capabilities of ICs, along with the high precision and throughput of microfluidics, in tandem with the specificity and sensitivity of biological systems for sensing and reporting on their environment. The resulting modularity could enhance robustness and expand the applicability of these devices, potentially driving notable advancements across diverse scientific fields.

In our initial classification framework, secure and unsecured biological fully integrated customized systems ((Sec)BioFICS) are the most represented, accounting for 38% of the 121 papers classified. By contrast, biological hybrid electrical laboratory-scale off-the-shelf systems (BioHELOS), biological hybrid electronic miniaturized off-the-shelf systems (BioHEMOS) and biological pure optical laboratory-scale off-the-shelf systems (BioPOLOS) are the least represented, each comprising 7–10% of those papers. When we examine only the past 3 years, this distribution has remained largely consistent, with a modest increase for BioHELOS and BioHEMOS, each now comprising 15% of

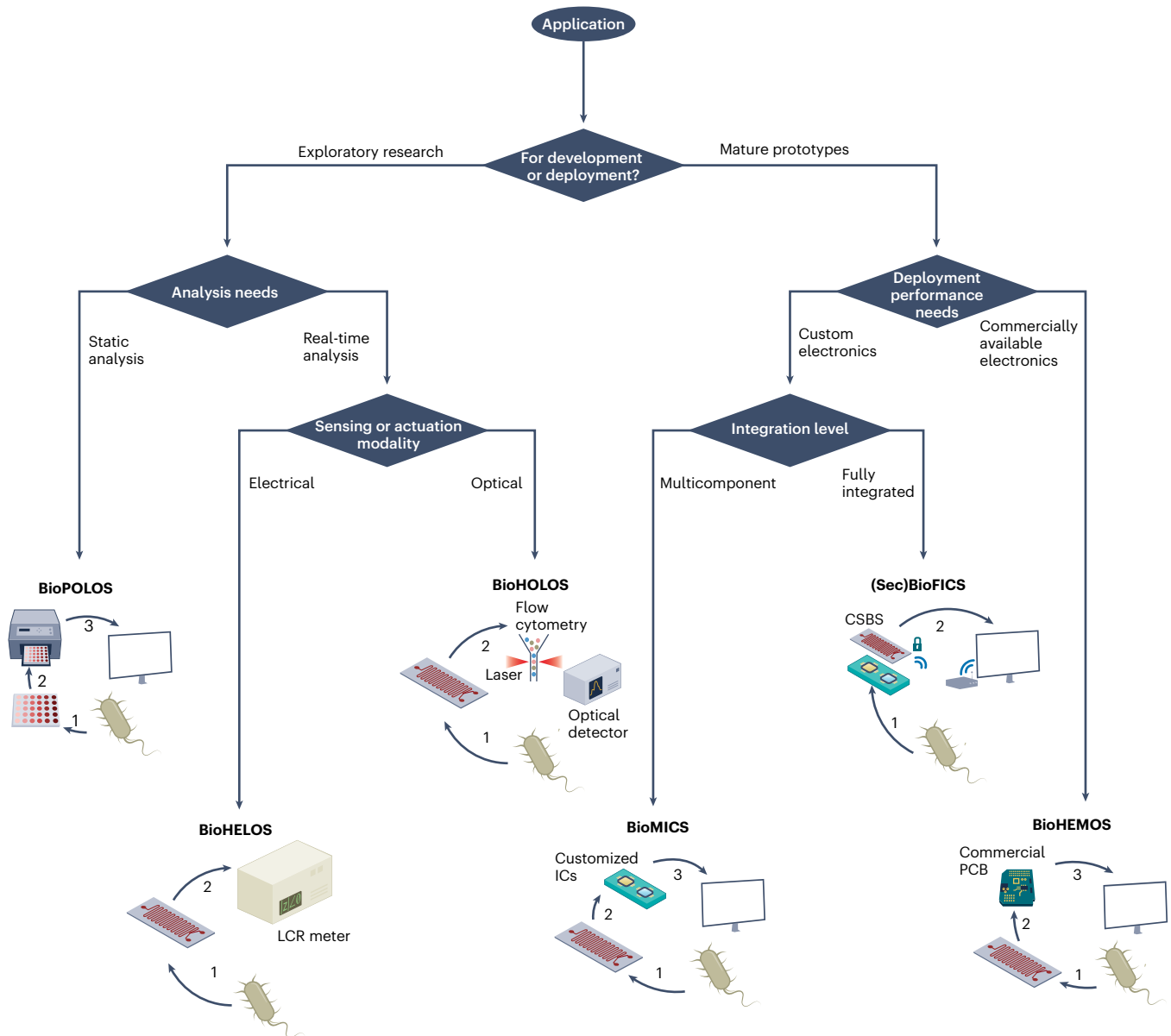


Fig. 6 | Decision diagram illustrating the design choices for various applications of engineered biological systems. The decision tree demonstrates the motivations behind design choices with an example and providing a classification framework. The intended use case and design maturity drive the first decision point, distinguishing between exploratory research conducted in a controlled laboratory environment and the deployment of mature prototypes in the field. In the development branch, systems are classified based on analysis and acquisition requirements: either static analysis of one-time acquired biological samples or real-time analysis of acquired biological samples. The application needs, budget and system constraints will determine which path will be chosen. Real-time analysis with microfluidic devices typically uses electrical or optical modalities to control and monitor engineered biological systems with high-precision benchtop equipment in a laboratory environment. The selection of these options will depend on the biological mechanism that is available and the user's need

for its precision, accuracy and speed. In the deployment branch, performance requirements guide the decision between using custom electronics tailored for optimized, application-specific performance and cost-effective, commercially available electronics designed for general performance. The selection of these options depends on the availability of components, packaging needs, cost and the expertise of the development group related to custom electronics design, development, fabrication and testing. Customized systems can be categorized by their level of integration and the required physical interaction for deployment. Multicomponent systems are typically designed for short-term deployment, whereas fully integrated systems are intended for minimal physical interaction, allowing for long-term deployment in remote, inaccessible areas, such as soil and wastewater. Furthermore, these fully integrated systems can be engineered to be cost-effective, making them also well suited for single-use applications, such as in-body monitoring via ingestible capsules.

the total. (Sec)BioFICS continues to lead at 37%, while BioPOLOS and biological multicomponent integrated customized systems (BioMICS) become the least represented at 7%. These classification statistics are accessible via the 'Living Roadmap' (<https://www.programmingbiology.org/csbs>), which is continuously updated to reflect the latest developments.

Our initial examination of 121 papers also shows that (Sec)BioFICS designs have steadily advanced in system integration and complexity.

By leveraging technology scaling and engineering innovations, these designs have achieved multiple functionalities within a compact footprint. This progress has substantially reduced the cost per functionality, enhancing efficiency and cost-effectiveness for practical applications. By contrast, the BioPOLOS, BioHELOS and biological hybrid optical laboratory-scale off-the-shelf system (BioHOLOS) categories, which rely on highly accurate benchtop electrical or optical laboratory equipment, have not undergone a comparable reduction

in cost per functionality, highlighting a gap in their scalability and economic feasibility.

Looking ahead, we expect biological developments to drive future trends more rapidly than advances in microfluidics or electronics. These developments could include the introduction of orthogonal signaling molecules²⁰⁵, new intercellular communication techniques^{206–208} or more predictive combinatorial design to create parallel sensing variants^{209–212}. Currently, the electronics field as an engineered discipline is highly mature²¹³. In most cases, the electronic capabilities are far more advanced than biology and most of the engineering effort is in getting the biology to work reliably²¹⁴. Electronics then must be able to detect noisy signals or nondeterministic behavior.

Biological advances will not only reduce the design effort and lead to the standardization of electronics but will also open a much wider set of application areas. We anticipate that BioHEMOS designs will become increasingly prevalent as the availability of off-the-shelf miniaturized electronics continues to expand. These advancements are expected to enhance scalability and drive cost reductions, making BioHEMOS designs more accessible and widely adopted. High-throughput designs may also become more attractive as the level of automation increases²¹⁵. In this case, the use of BioPOLOS systems may decrease due to their low integration level and high costs. In all cases, getting more classifications used or balancing the distribution of classifications is not the goal of the field. Rather, the goal is to have the tools available to achieve the application requirements so that they meet performance and economic needs.

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Competing interests

The authors declare no competing interests.

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